MET 359 Assignment

Climate Variability and Trends in Uganda Using CRU TS4.07 Data

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Abstract

This study analyzed the spatial distribution, temporal trends, and decadal variability of monthly precipitation over Uganda for the period 1990-2020 using the CRU TS4.07(Harris & Jones, 2022) gridded dataset. The mean precipitation map revealed a distinct spatial pattern, with the wettest conditions concentrated in the south-central region around the Lake Victoria basin (longitudes 31–33°E, latitudes 0° to −1°S), recording mean values exceeding 160 mm/month, while the northeastern areas remained comparatively drier at approximately 40 mm/month. Temporal analysis of the spatially averaged annual mean precipitation indicated a period mean of 106.18 mm/month with a standard deviation of 9.10 mm/month and a statistically positive linear trend of +0.551 mm/month per year, suggesting a gradual increase in precipitation over the study period. Decadal comparison between 1990–1999 and 2010–2020 revealed a predominantly wetter signal across the eastern and central regions, with the largest increases concentrated near the Lake Victoria basin, while northwestern Uganda experienced modest drying. Hotspot analysis identified the top five areas of greatest increase in the southeastern region and the top five areas of greatest decrease in the northwestern region. These findings highlight the spatially heterogeneous nature of precipitation change over Uganda and underscore the importance of regional climate monitoring for water resource management and agricultural planning.

Data and Methods

1. Dataset

Monthly precipitation data were obtained from the CRU TS4.07 dataset, produced by the University of East Anglia (Harris & Jones, 2022). CRU TS4.07 provides high-resolution gridded climate data at a spatial resolution of 0.5° × 0.5° latitude-longitude, derived from interpolation of station observations across the global land surface. The variable of interest was monthly precipitation (pre), extracted for the period January 1990 to December 2020, yielding a total of 360 monthly time steps.

2. Bounding Box Definition

The spatial domain of the analysis was defined by a bounding box adequately covering Uganda's territorial extent, as specified in Table 1 below:

Boundary	Value
Minimum Latitude	−1.5°N
Maximum Latitude	4.5°N
Minimum Longitude	29.5°E
Maximum Longitude	35.0°E

All analyses and visualizations were restricted to grid cells falling within this bounding box.

3. Data Processing and Analysis

Data was loaded and processed using Python (version 3), with the following libraries:

- **xarray** — for loading, subsetting, and computing statistics on the NetCDF dataset
- **numpy** — for numerical computations, including trend fitting and variability statistics
- **matplotlib** — for all visualizations

Results

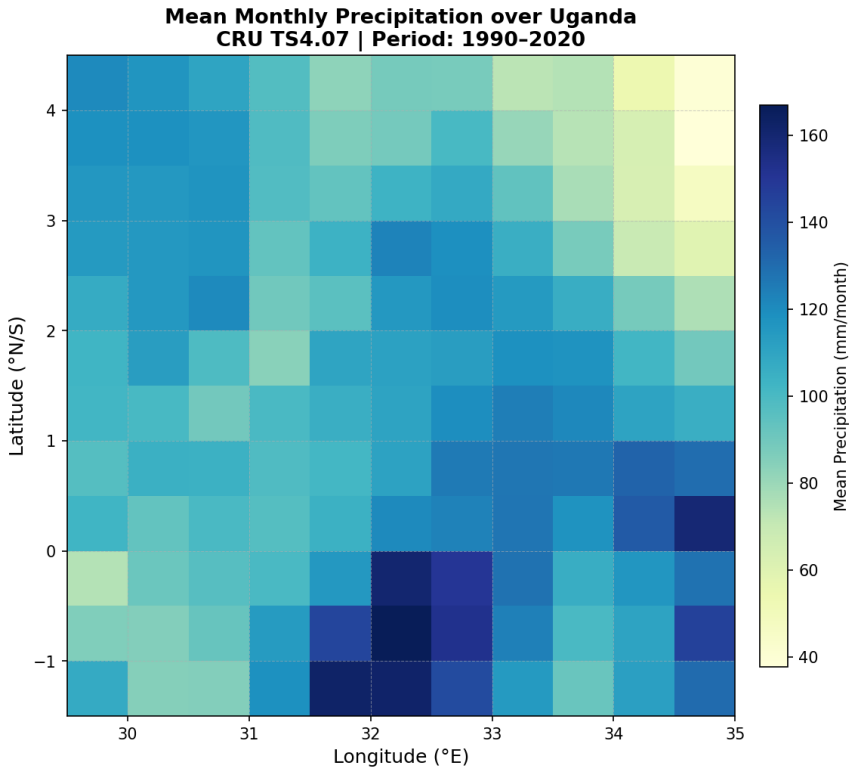

Figure 1: Mean monthly precipitation over Uganda (1990–2020), CRU TS4.07.

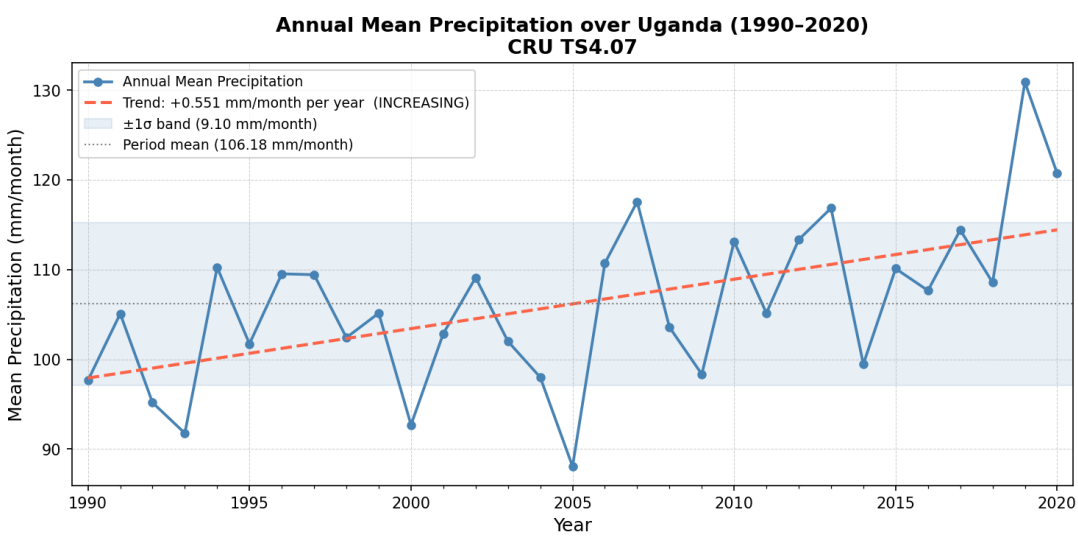

Figure 2: Annual mean precipitation time series over Uganda (1990–2020) with linear trend and ±1σ variability band, CRU TS4.07.

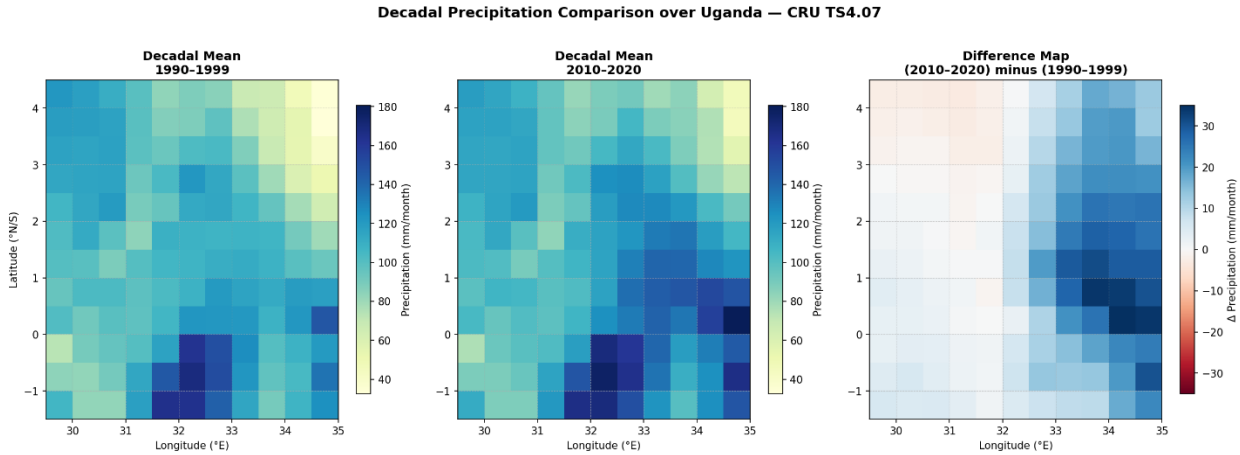

Figure 3: Decadal mean precipitation for 1990–1999 and 2010–2020, and difference map (2010–2020 minus 1990–1999), CRU TS4.07.

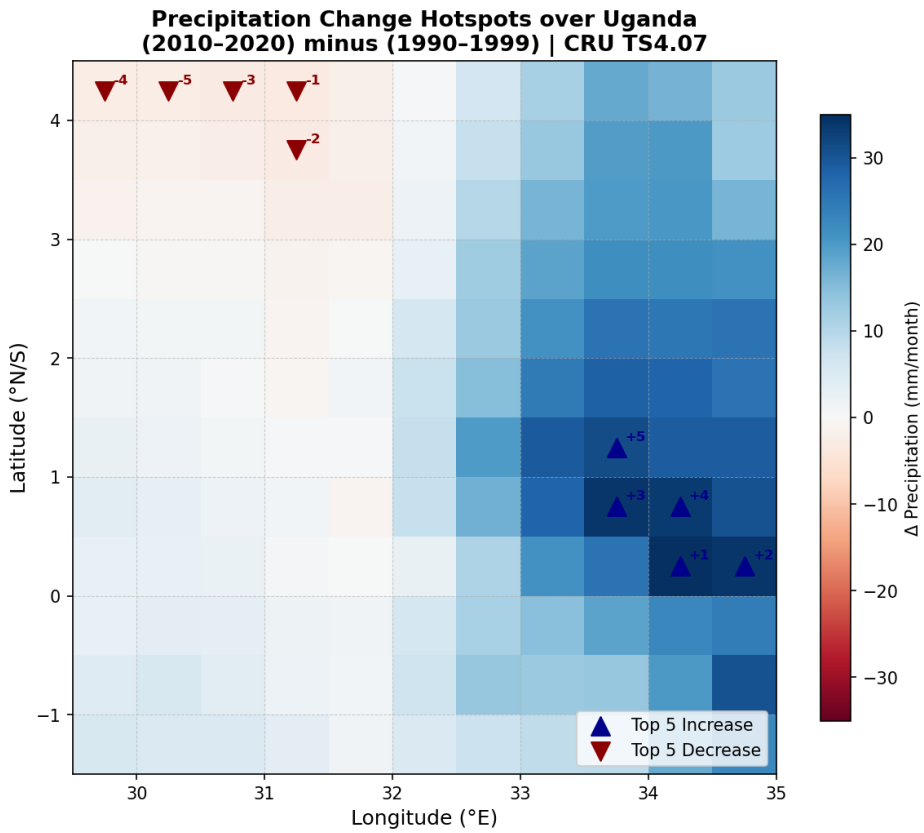

Figure 4: Precipitation change hotspots over Uganda showing top 5 grid points of largest increase (blue triangles) and largest decrease (red triangles), CRU TS4.07.

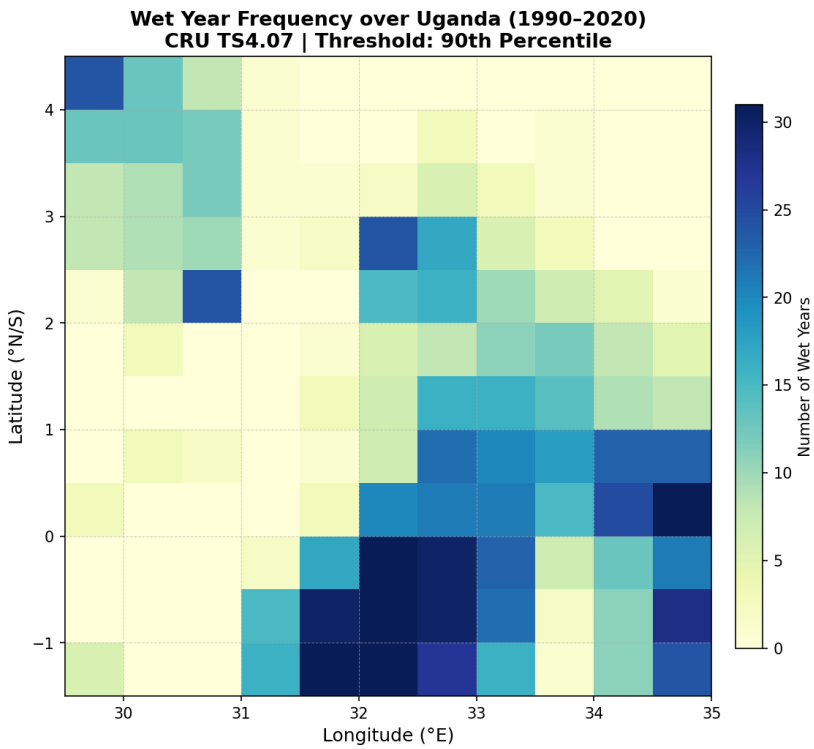

Figure 5: Spatial distribution of wet year frequency over Uganda (1990–2020), defined as years exceeding the 90th percentile threshold. CRU TS4.07.

1. **Spatial Distribution of Mean Precipitation**

The spatial distribution of mean monthly precipitation over Uganda for the period 1990–2020 is shown in Figure 1. Precipitation exhibited a clear spatial gradient across the country, with the wettest conditions concentrated in the south-central region around the Lake Victoria basin, particularly between longitudes 31–33°E and latitudes 0° to −1°S, where mean monthly values exceeded 160 mm/month. Central Uganda recorded moderate precipitation values ranging between 100–130 mm/month. The driest conditions were observed in the northeastern corner of the bounding box, approximately around latitude 4°N and longitude 34–35°E, with values as low as 40 mm/month. This spatial pattern reflects the strong influence of Lake Victoria on moisture availability in southern Uganda, as well as the increasing aridity towards the northern and northeastern parts of the country.

2. **Annual Mean Precipitation Time Series and Trend**

The annual mean precipitation time series for Uganda over 1990–2020 is presented in Figure 2. The mean was 106.18 mm/month with a standard deviation of 9.10 mm/month, indicating

moderate interannual variability. The coefficient of variation further confirms that annual precipitation fluctuated meaningfully around the long-term mean throughout the study period.
A linear trend of +0.551 mm/month per year was identified, indicating a gradual but consistent increase in precipitation over the 31 years. The most notable dry year was 2005, when annual mean precipitation fell below 90 mm/month, falling outside the ±1σ band. Conversely, 2019 recorded the highest annual mean of approximately 131 mm/month, also exceeding the upper boundary of the variability band. The general upward trend suggests that Uganda has experienced a modest but progressive increase in rainfall since 1990, though considerable year-to-year variability persists.

3. **Decadal Precipitation Comparison and Difference Map**

Figure 3 shows the decadal mean precipitation maps for 1990–1999 and 2010–2020, alongside the difference map, showing spatial changes between the two periods. Both decadal maps exhibit a broadly similar spatial pattern, with the Lake Victoria basin remaining the wettest region in both periods. However, the difference map reveals important spatial shifts in precipitation between the two decades. The eastern and south-central regions, particularly around longitudes 33–35°E, experienced a notable increase in precipitation in the more recent decade, with positive anomalies reaching approximately +30 mm/month in some grid cells. In contrast, the northwestern region of Uganda, around longitudes 29–31°E and latitudes above 3°N, shows a modest decline in precipitation between the two periods, with negative anomalies of approximately −10 mm/month. These findings suggest that while Uganda has become wetter on average, this trend is spatially uneven, with the east getting wetter and the northwest experiencing some drying.

4. **Precipitation Change Hotspots**

The spatial distribution of precipitation change hotspots is shown in Figure 4. The top 5 grid points with the largest increases in precipitation were concentrated in the southeastern region of Uganda, clustered around the Lake Victoria basin between latitudes 0°–1.5°N and longitudes 33–35°E. The top 5 grid points with the largest decreases were located in the northwestern region, clustered near latitudes 3.75°–4.25°N and longitudes 29.5°–31.5°E. This spatial contrast reinforces the pattern identified in the difference map, confirming a clear east-west dipole in precipitation change over Uganda during the study period, with the southeast becoming progressively wetter and the northwest experiencing drying.

Table 2.1: Top 5 Grid Points with Largest Increase in Precipitation (2010–2020 vs 1990–1999)

Rank	Latitude (°N)	Longitude (°E)	ΔPrecipitation (mm/year)
1	0.25	34.25	+417.76
2	0.25	34.75	+408.19
3	0.75	33.75	+407.64
4	0.75	34.25	+402.77
5	1.25	33.75	+376.97

Table 2.2: Top 5 Grid Points with Largest Increase in Precipitation (2010–2020 vs 1990–1999)

Rank	Latitude (°N)	Longitude (°E)	ΔPrecipitation (mm/year)
1	4.25	31.25	−37.18
2	3.75	31.25	−37.18
3	4.25	30.75	−33.26
4	4.25	29.75	−32.28
5	4.25	30.25	−31.41

The increase values (+400 mm/year) seem very large compared to the decrease values (−37 mm/year). This suggests the southeastern region around Lake Victoria experienced a dramatic intensification of precipitation between the two decades, while the northwestern drying was comparatively modest.

5. **Wet Year Frequency**

Figure 5 presents the spatial distribution of wet year frequency over Uganda for the period 1990–2020, where a wet year is defined as any year in which annual mean precipitation exceeded the 90th percentile threshold of the spatially averaged time series. The map reveals considerable spatial variability in wet year occurrence across the country. The highest frequency of wet years was recorded in the south-central region around the Lake Victoria basin, particularly between longitudes 31–33°E and latitudes 0° to −1°S, where some grid cells recorded up to 31 wet years out of the 31-year study period, indicating that this region consistently experiences above-threshold precipitation. A secondary cluster of high wet year frequency was also observed around latitude 3°N and longitude 32°E in central Uganda. In contrast, the northeastern and western parts of the bounding box recorded the fewest wet years, with several grid cells showing zero occurrences, reinforcing the pattern of relatively drier conditions in these areas identified in earlier analyses. Overall, the spatial pattern of wet year frequency closely mirrors the mean precipitation distribution shown in Figure 1, confirming that wetter regions not only receive more rainfall on average but also experience extreme wet conditions more frequently.

Conclusion

This study examined precipitation variability and trends over Uganda for the period 1990–2020 using the CRU TS4.07 gridded dataset (Harris & Jones, 2022). Results confirmed a distinct spatial gradient in mean precipitation, with the Lake Victoria basin consistently recording the highest rainfall, exceeding 160 mm/month, while northeastern Uganda remained the driest region at approximately 40 mm/month. A positive linear trend of +0.551 mm/month per year was identified, indicating a gradual increase in precipitation over the study period, though considerable interannual variability persisted with a standard deviation of 9.10 mm/month. Decadal comparison revealed a spatially uneven wetting signal, with the eastern and south-central regions becoming progressively wetter and the northwest experiencing modest drying between the 1990s and 2010s. Wet year frequency analysis further confirmed the Lake Victoria basin as the most precipitation-extreme region in Uganda. These findings highlight the spatially heterogeneous nature of precipitation change over Uganda and have direct implications for water resource management and agricultural planning in the region.

References

Harris, I. C., & Jones, P. D. (2022). CRU TS4.07: Climatic Research Unit (CRU) Time-Series (TS) Version 4.07 of High-Resolution Gridded Data of Month-by-month Variation in Climate (January 1901-December 2022). Centre for Environmental Data Analysis. https://crudata.uea.ac.uk/cru/data/hrg/